

Experiment No. 5

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Experiment Title

Apply array methods and object handling; create a cart total calculator with discount logic.

Software / Tools Required

- Visual Studio Code
- Google Chrome / Microsoft Edge
- HTML5
- CSS3
- JavaScript

Experiment Program Code

Files Used:

- index.html
- style.css
- script.js

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Shopping Cart Calculator</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
  <h1>■ Shopping Cart Calculator</h1>

  <div class="add-item">
    <input type="text" id="name" placeholder="Product Name">
    <input type="number" id="price" placeholder="Price" min="0">
    <input type="number" id="qty" placeholder="Qty" value="1" min="1">
    <select id="discount">
      <option value="0">Discount 0%</option>
      <option value="5">Discount 5%</option>
      <option value="10">Discount 10%</option>
      <option value="15">Discount 15%</option>
      <option value="20">Discount 20%</option>
    </select>
    <button id="addBtn">Add Item</button>
  </div>

  <table>
    <thead>
      <tr>
        <th>Product</th><th>Price</th><th>Qty</th>
        <th>Total</th><th>Action</th>
      </tr>
    </thead>
    <tbody id="cartBody"></tbody>
  </table>

  <div class="summary">
    <p>Subtotal: <span id="subtotal">■0.00</span></p>
    <p>Discount: <span id="discountAmount">■0.00</span></p>
    <p>Tax (10%): <span id="tax">■0.00</span></p>
    <h2>Grand Total: <span id="grand">■0.00</span></h2>
  </div>
</div>
<script src="script.js"></script>
</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  font-family: Arial, Helvetica, sans-serif;
}

body {
  background: #f2f5f9;
  display: flex;
  justify-content: center;
  align-items: flex-start;
  padding: 40px;
}

.container {
  width: 1000px;
  background: white;
  padding: 25px;
  border-radius: 12px;
  box-shadow: 0 10px 25px rgba(0,0,0,.15);
}

h1 {
  text-align: center;
  margin-bottom: 25px;
  color: #333;
}

.add-item {
  display: grid;
  grid-template-columns: 2fr 1fr 1fr 1.3fr auto;
  gap: 10px;
  margin-bottom: 25px;
}

.add-item input, .add-item select {
  padding: 12px;
  border: 1px solid #ccc;
  border-radius: 6px;
}

button {
  background: #28a745;
  color: white;
  border: none;
  padding: 12px 20px;
  border-radius: 6px;
  cursor: pointer;
}

button:hover { background: #218838; }

table {
  width: 100%;
  border-collapse: collapse;
  margin-bottom: 25px;
}

table th {
  background: #007bff;
  color: white;
  padding: 12px;
}

table td {
  text-align: center;
  padding: 12px;
  border-bottom: 1px solid #ddd;
}

.remove { background: #dc3545; }
.remove:hover { background: #b52a37; }

.summary {
  width: 300px;
  margin-left: auto;
  text-align: right;
}

.summary p, .summary h2 { margin: 10px 0; }
.summary h2 { color: #007bff; }

@media (max-width: 768px) {
  .container { width: 100%; }
  .add-item { grid-template-columns: 1fr; }
  .summary { width: 100%; text-align: center; }
}
```

script.js

```
const addBtn = document.getElementById("addBtn");
const cartBody = document.getElementById("cartBody");

let cart = [];

addBtn.addEventListener("click", () => {
  const name = document.getElementById("name").value.trim();
  const price = parseFloat(document.getElementById("price").value);
  const qty = parseInt(document.getElementById("qty").value);
  const discount = parseFloat(document.getElementById("discount").value);

  if (name === "" || isNaN(price) || price < 0 ||
      isNaN(qty) || qty < 1) {
    alert("Please enter valid product details.");
    return;
  }

  cart.push({ name, price, qty, discount });

  document.getElementById("name").value = "";
  document.getElementById("price").value = "";
  document.getElementById("qty").value = 1;

  displayCart();
});

function displayCart() {
  cartBody.innerHTML = "";
  let subtotal = 0;
  let discountAmount = 0;

  cart.forEach((item, index) => {
    const total = item.price * item.qty;
    subtotal += total;
    discountAmount += (total * item.discount) / 100;

    cartBody.innerHTML += `
    <tr>
      <td>${item.name}</td>
      <td>■${item.price.toFixed(2)}</td>
      <td>${item.qty}</td>
      <td>■${total.toFixed(2)}</td>
      <td>
        <button class="remove" onclick="removeItem(${index})">
          Remove
        </button>
      </td>
    </tr>`;
  });

  const amountAfterDiscount = subtotal - discountAmount;
  const tax = amountAfterDiscount * 0.10;
  const grand = amountAfterDiscount + tax;

  document.getElementById("subtotal").textContent =
    `■${subtotal.toFixed(2)}`;
  document.getElementById("discountAmount").textContent =
    `■${discountAmount.toFixed(2)}`;
  document.getElementById("tax").textContent =
    `■${tax.toFixed(2)}`;
  document.getElementById("grand").textContent =
    `■${grand.toFixed(2)}`;
}

function removeItem(index) {
  cart.splice(index, 1);
  displayCart();
}
```

Application Description

The application accepts product name, price, quantity, and discount percentage. Products are stored as objects in an array. JavaScript calculates the subtotal, discount amount, 10% tax, and grand total. The user can also remove items from the cart.

Features Implemented

Feature	Description
Product Entry	Accepts product name, price, quantity, and discount.
Array Handling	Stores cart items in an array.
Object Handling	Each cart item is represented as an object.
Discount Logic	Calculates discount amount for every item.

Feature	Description
Cart Calculation	Computes subtotal, tax, and grand total.
DOM Manipulation	Updates the cart table and summary dynamically.
Event Handling	Add Item and Remove buttons execute JavaScript.

Output

Shopping Cart Calculator

Product	Price	Qty	Total	Action
Notebook	₹120.00	2	₹240.00	Remove
Pen Set	₹80.00	3	₹240.00	Remove

Subtotal:

₹480.00

Discount:

₹36.00

Tax (10%):

₹44.40

Grand Total:

₹488.40

Output Summary

- Accepts product details through input fields.
- Stores products as objects in an array.
- Displays product-wise totals in a table.
- Applies the selected discount.
- Calculates 10% tax and grand total.
- Allows cart items to be removed.

Result / Conclusion

The Shopping Cart Calculator was implemented successfully using HTML, CSS, and JavaScript. The experiment demonstrated array handling, object handling, event handling, DOM manipulation, discount calculation, tax calculation, and dynamic cart updates.